

Bridging the translational gap in Alzheimer's disease: A bibliometric and evidence-based assessment of secretome and stem cell therapeutics

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ABSTRACT

Introduction: Alzheimer's disease (AD) is a significant worldwide challenge where few disease-modifying treatments are available. Additionally, secretome- and stem cell-based therapeutics have emerged as potential translational strategies. However, the research landscape has not been systematically assessed.

Methods: A bibliometric analysis (2003–2025) was performed using VOSviewer and Biblioshiny from Scopus data. Research hotspots, knowledge gaps, and translational implications were summarized. Preclinical and clinical evidence were combined for context.

Results: There were 182 papers identified. Most of the funding was provided by the United States and China, reflecting different priorities for clinical translation and molecular innovation. Four themes were identified by co-occurrence analysis: (1) secretome and extracellular vesicles; (2) microglia and neuroinflammation; (3) aging and neurodegenerative diseases; and (4) neural stem cells and functional recovery. Publication output increased after 2016, with a growing trend towards the translation of regenerative and cell-free therapies.

Discussion: Secretomes and stem cell-based therapies offer compelling prospects for precision AD therapy; however, roadblocks to clinical translation remain, including standardization, delivery, and regulatory approval. Cross-disciplinary interaction and streamlined pipelines are critically required to address the translational bottleneck.

1. Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder primarily characterized by the accumulation of amyloid-beta plaques and tau protein tangles, which collectively lead to synaptic failure and neuronal death. AD represents a significant worldwide challenge, currently associated with an estimated 34–47 million cases, a prevalence expected to surpass 131 million by 2050 (Twiss et al., 2025). Beyond its biological impact, AD is deeply linked to economic and social fear and stigma in different cultural contexts (Giannouli, 2017). Often

conceptualized as a "modern plague," the disease is culturally identified with concepts of "Evil" or divine punishment, leading to pervasive social stigma for both patients and their families (Makri, 2022). Patients with early symptoms frequently fear being categorized with the AD diagnostic group due to the social weight of the diagnosis. Furthermore, the substantial economic costs, estimated at US\$ 1.31 trillion in 2019 and projected to reach US\$ 9.9 trillion by 2050, are largely driven by informal care (Kalochristianaki et al., 2025). This reliance on family support often occurs in a context where caregivers feel a wrong but "imposed requirement of hiding" the patient to manage a "spoiled

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identity," leading to social isolation and burnout. Currently approved medications, including cholinesterase inhibitors and memantine, offer only minor symptomatic benefits but do not affect disease progression (Zhang et al., 2020), underscoring the need to develop new disease-modifying therapeutic approaches.

In recent years, AD research has shown promise in the use of secretomes derived from stem cells as a potential neuroregenerative therapeutic. Secretomes, which are heterogeneous in the proteins, cytokines, and exosomes they secrete, have neuroprotective and anti-inflammatory activities (Mehrabadi, 2025). The preclinical studies have reported an augmentation of neurogenesis and synaptic function and a decrease in amyloid-beta ($A\beta$) deposition after treatment with secretomes from mesenchymal stem cells (MSCs) together with repression in the levels of inflammatory mediators such as IL-1 β , TNF- α , but elevation of neurotrophic factors, e.g., BDNF, NGF (Bali et al., 2017; Liu et al., 2020). However, the clinical use is limited by the poor capacity to obtain GMP-compliant, consistent, and standardized manufacturing (Zhang et al., 2014).

Neural stem cells (NSCs) and induced pluripotent stem cells (iPSCs) are considered potential clinical therapies for AD, as both possess the capacity to differentiate into functional neurons that can repair impaired neural circuits (Wang et al., 2019). These cells may be genetically modified to release neuroprotective molecules, such as insulin-like growth factor-1 (IGF-1), ultimately increasing their regenerative capability (Kang et al., 2016). It has been demonstrated that NSCs can promote neurogenesis and synaptic plasticity, with the potential to alleviate the cognitive deficits associated with AD (Mehrabadi, 2025). On the other hand, iPSCs offer an autologous, renewable cell source compared to allogeneic sources, resulting in minimal immunological rejection following personalized treatment approaches (Mehrabadi, 2025). Nevertheless, the clinical application of these cell-based strategies remains limited, and a comprehensive study of their mechanisms is necessary.

Cellular and cell-free therapy methods may have therapeutic effects on the major neuropathological hallmarks of AD, including reducing amyloid pathology and gliosis, increasing neuronal density, and improving cognitive function (Hambarsari et al., 2024). For instance, CM from BUMSCs has been reported to increase levels of neurotrophic factors (BDNF and NGF), promote neuroplasticity, and reduce neuroinflammation (Bali et al., 2017). Furthermore, sustained-release methods, such as intranasal application, demonstrate long-term effects on memory restoration and neuropathological amelioration in experimental models (Kang et al., 2016). Currently, the significant barriers include determining the effectiveness of promising biomarkers, optimizing therapeutic strategies, and facilitating large-scale production for clinical use (Wang et al., 2019).

We conducted a bibliometric analysis of secretomes derived from stem cells for the treatment of Alzheimer's disease. While the secretome approach has proven to be an interesting multitarget therapeutic strategy, its clinical use remains limited by standardization and scalability (Zhang et al., 2020). No bibliometric study has been conducted to summarize the progress, themes, and characteristics of this field. The purpose of this study is to analyze publication trends, research hotspots, and knowledge gaps, thereby reflecting the current state of scientific knowledge and providing insights for integrating experimental results into clinical practice.

2. Methods

2.1. Data collection and screening

A systematic literature search was conducted on June 27, 2025, using the Scopus database to identify articles relevant to Alzheimer's disease, the secretome, and stem cell-based therapy. The search strategy focused on titles, abstracts, and keywords using the following query: TITLE-ABS-KEY ("Alzheimer's disease" OR "AD") AND ("secretome" OR

"stem cell therapy") AND ("clinical application" OR "therapeutic target"). The time frame for inclusion (2003–2025) was selected to capture the early development and evolution of stem cell-based and secretome-related research in Alzheimer's disease. The early 2000s marked the initial exploration of stem cell applications in neurodegenerative disorders, while the concept of secretome and extracellular vesicles gained significant attention after approximately 2010. Therefore, this period enables a comprehensive longitudinal analysis of both foundation and translational advances in the field. The initial search identified 533 records, within the predefined time frame (2003–2025). To filter the dataset, the inclusion criteria included peer-reviewed journal articles in English, and at the final stage of publication. Duplicates, review papers, and conference proceedings were excluded. Overall, 182 articles met the eligibility criteria after screening and were selected for analysis (Fig. 1). Scopus was selected due to its extensive coverage of peer-reviewed literature across biomedical and interdisciplinary fields, as well as its provision of standardized and comprehensive metadata (e.g., author affiliations, citations, and keywords), which are essential for robust bibliometric analyses using tools such as VOSviewer and Biblioshiny.

2.2. Data extraction

Finally, 182 full papers that meet the publication criteria were selected and exported from Scopus into CSV format. This extraction is a valuable bibliographic dataset that provides necessary information, including article title, author(s) name(s), year of publication, journal title or name, keywords, and abstract. This well-compiled dataset lays the groundwork for further bibliometric analysis and visualization to explore the trends in research on Alzheimer's disease, the secretome, and stem cell therapy.

2.3. Data analysis

Data were analyzed using Scopus Analyze Results, Biblioshiny, and VOSviewer (version 1.6.20). These analytical software instruments facilitate the development of a bibliometric map and network view, including co-authorship, co-citation, and keyword co-occurrence analysis. Using Biblioshiny and VOSviewer, this study provides a visual representation of interactions among authors, institutions, keywords, and publications, revealing the major research trends and identifying who drove them in this field. Commonly used keywords by the authors were employed throughout the analysis to maintain consistency when identifying thematic clusters and scientific focus.

3. Results

3.1. Analysis of publication growth

The studies included in the analysis covered the period from 2003 to 2025 (Fig. 2). As shown in Fig. 2, publication output was initially low, with only one paper published per year between 2003 and 2004, and remained limited until 2010. From 2011 onward, the number of publications increased steadily, peaking at 29 in 2024 before declining slightly to 9 in 2025. These findings indicate a clear upward trend in research activity and growing interest in the topic over the past decade.

Fig. 2 also illustrates the trend in citation impact over time. Despite the increase in the number of publications, the mean citations per article (MeanTCperArt) showed substantial variability. The highest values were observed in 2010 (278.25) and 2009 (191.5), indicating that studies published during these years had particularly high impact. In contrast, more recent publications in 2024 and 2025 had considerably lower mean citation values (16.97 and 0.78, respectively). This decline is likely due to the shorter time available for newer articles to accumulate citations.

Furthermore, the MeanTCperYear reached its highest levels in 2010

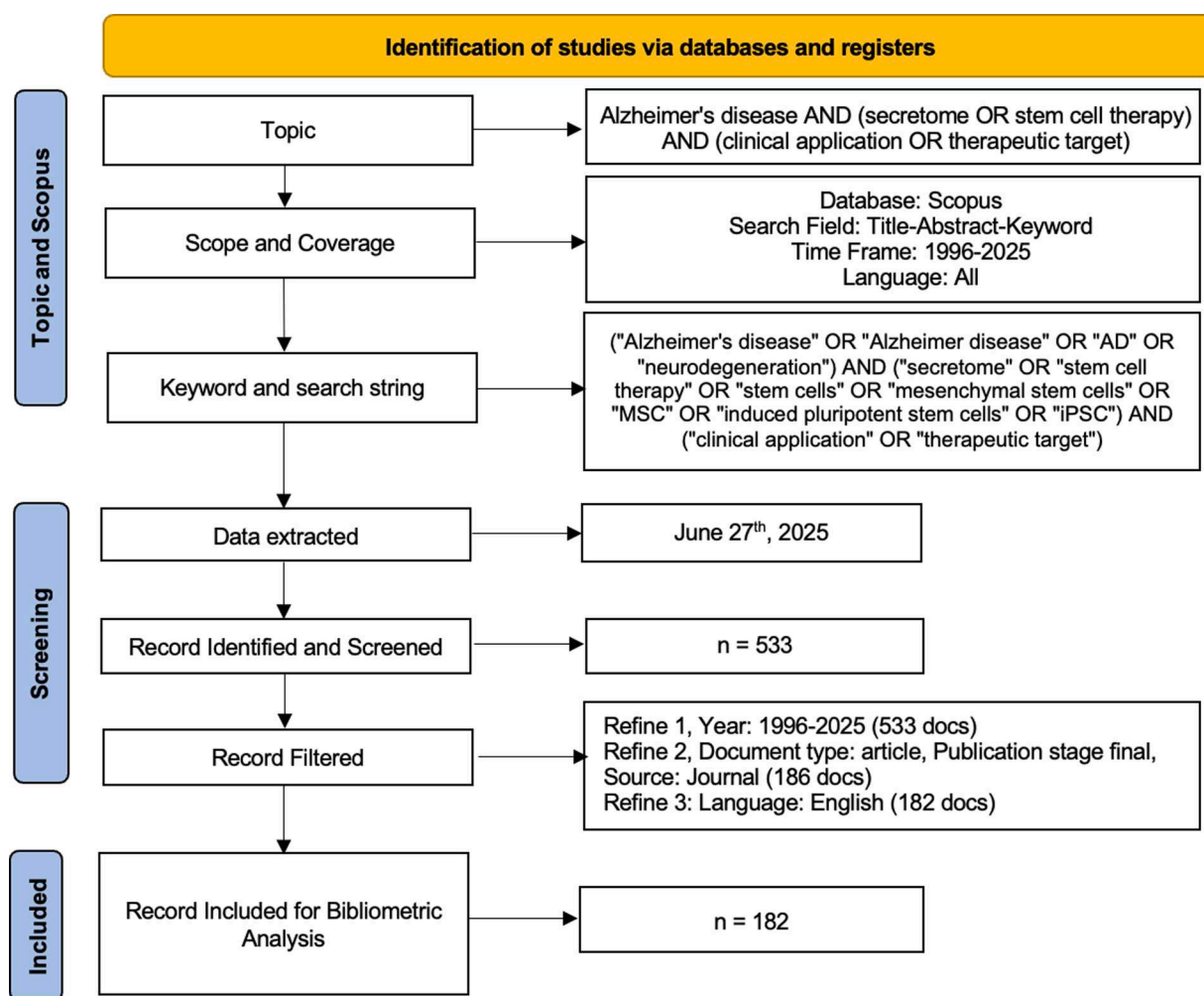


Fig. 1. Study identification.

(17.39) and 2023 (12.54), suggesting that certain years produced especially influential studies. Overall, these results demonstrate that while publication output has increased over time, citation impact is not solely determined by the number of publications but is also influenced by factors such as research novelty, methodological quality, and scientific visibility.

3.2. Analysis of countries

Single-country publications (SCP) refer to studies conducted within a single country, whereas multiple-country publications (MCP) involve international collaboration. As shown in Fig. 3, the United States produced the highest number of publications ($n = 51$), the majority of which were domestic (79.2% SCP), with a smaller proportion involving international collaboration (20.8% MCP). China ranked second with 32 publications but demonstrated a higher level of international collaboration, with 31.2% of its output classified as MCP. The United Kingdom and Japan exhibited moderate and relatively balanced levels of international engagement, with MCP proportions of 25% and 30%, respectively. In contrast, several countries, such as South Korea, Italy, and Malaysia, contributed exclusively through SCPs, indicating a stronger focus on nationally conducted research. Fig. 3 further illustrates that some countries were more actively involved in international collaborations, such as Germany (66.7% MCP) and India (50% MCP), both of which had substantial proportions of internationally co-authored studies. Conversely, countries including Belgium, Austria, Croatia, Ireland, and Poland contributed exclusively through MCPs, indicating

full participation in international collaborative research.

3.3. Analysis of authors

A total of one thousand one hundred fifty three authors have contributed to this research field. As shown in Fig. 4, Wang S and Zhang Y are the most productive authors, each publishing six papers between 2018 and 2025. Wang S primarily focuses on cellular signaling pathways and mechanisms related to Alzheimer's disease and neurodegeneration, whereas Zhang Y specializes in stem cell-based regenerative therapies for Alzheimer's disease. Other prominent contributors include Bennett D.A. ($n = 5$), Wang X ($n = 5$), Chen X ($n = 4$), Hargus G ($n = 4$), Canoll P ($n = 3$), De Jager P.L. ($n = 3$), Lee H ($n = 3$), and Li J ($n = 3$). Fig. 4 further indicates that the majority of these high-output authors are affiliated with institutions in China, while Bennett D.A., Hargus G., Canoll P., and De Jager P.L. are based in the United States, and Lee H is affiliated with South Korea.

3.4. Analysis of journals

A total of 11 journals contributed to publications in this research area. As shown in Fig. 5, Responses ($n = 16$) and International Journal of Molecular Sciences (*IJMS*) ($n = 8$) were the most productive journals, primarily focusing on molecular mechanisms and regenerative strategies, followed by *Cells* ($n = 7$). Neural Regeneration Research and Stem Cell Research & Therapy each published four articles, reflecting sustained attention to neural regeneration topics. Fig. 5 also indicates that

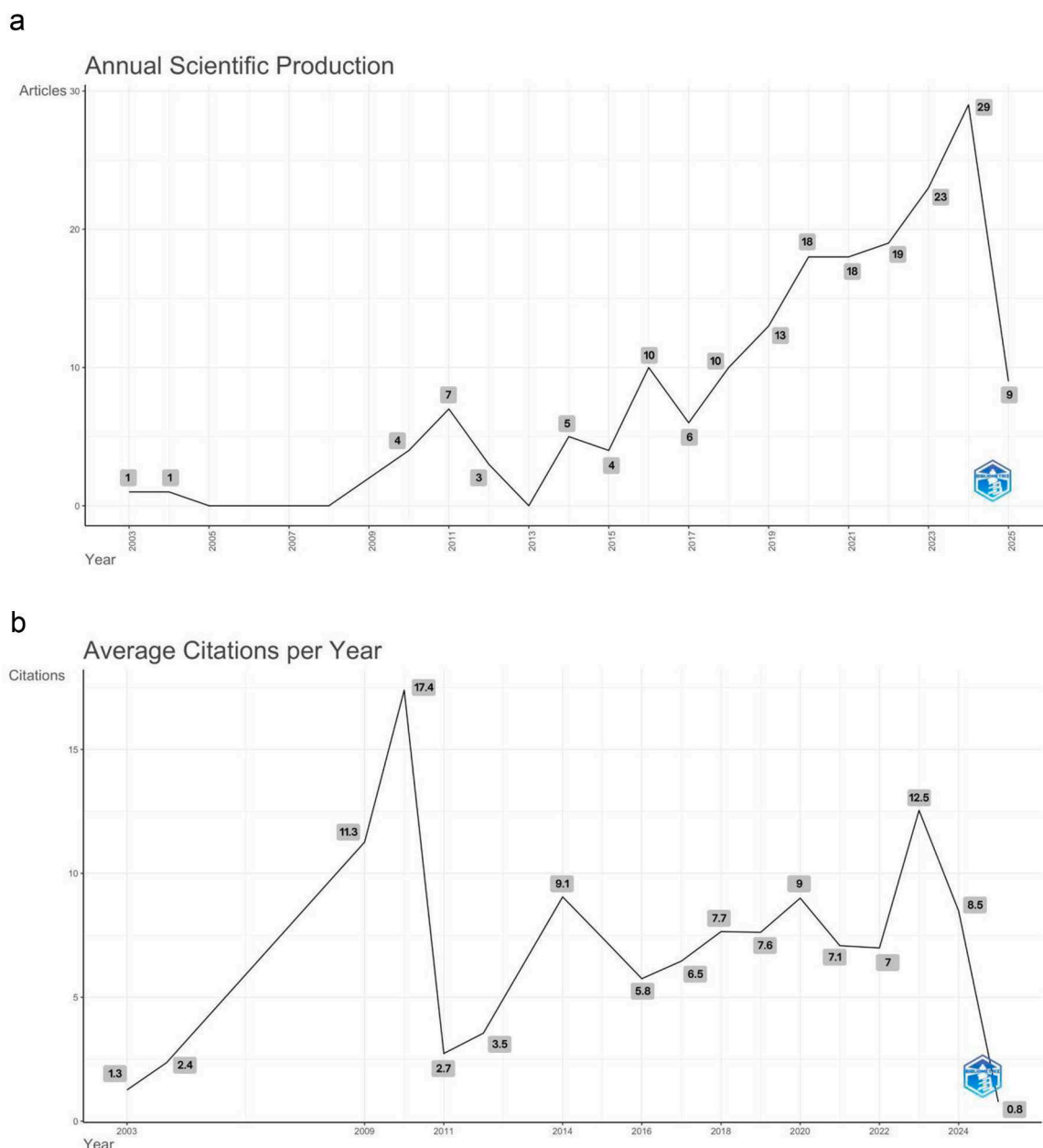


Fig. 2. Analysis of publication growth.

other influential journals, including *Frontiers in Aging Neuroscience*, *Neurobiology of Aging*, and *Stem Cell Reports*, contributed to the field by integrating perspectives from neurodegeneration, stem cell science, and aging research. Overall, these findings highlight the increasingly interdisciplinary and evolving scientific landscape of this research area.

3.5. Keywords and topic trends

Most studies in this dataset were on Alzheimer's disease and its neurodegenerative mechanism. The most dominant keywords were Alzheimer's disease, neurodegeneration, neuroinflammation, and microglia (Fig. 6), highlighting the field's primary focus on disease pathology, immune signaling, and glial activity. Such terms as "stem cells," "extracellular vesicles," "apoptosis," and regenerative medicine" also showed the emerging application of cell- and molecular-based therapy for neurodegenerative diseases.

The keywords, as depicted in Fig. 6, change over time, revealing that past research has focused more on specific topics, such as microglia,

neuroinflammation, and biomarkers, which remain important. The hot topics in recent years have been "extracellular vesicles," "regenerative medicine," and "stem cell therapy." It shows a shift from mechanistic to translational and regenerative applications, especially during the pre-2023 period.

3.6. Cluster analysis

Four thematic clusters were identified in the co-occurrence analysis (Fig. 7), reflecting the dominance of AD research within the stem cell field. As shown in Fig. 7A, the red cluster links AD with stem cell therapy and extracellular vesicles, highlighting the growing emphasis on regenerative medicine approaches. The green cluster ($n = 20$) is associated with immune-related processes, including neuroinflammation, microglial function, and apoptosis, suggesting that inflammatory mechanisms may represent important therapeutic targets. The yellow cluster captures the intersection between AD and other neurodegenerative diseases, such as Parkinson's disease (PD) and amyotrophic lateral

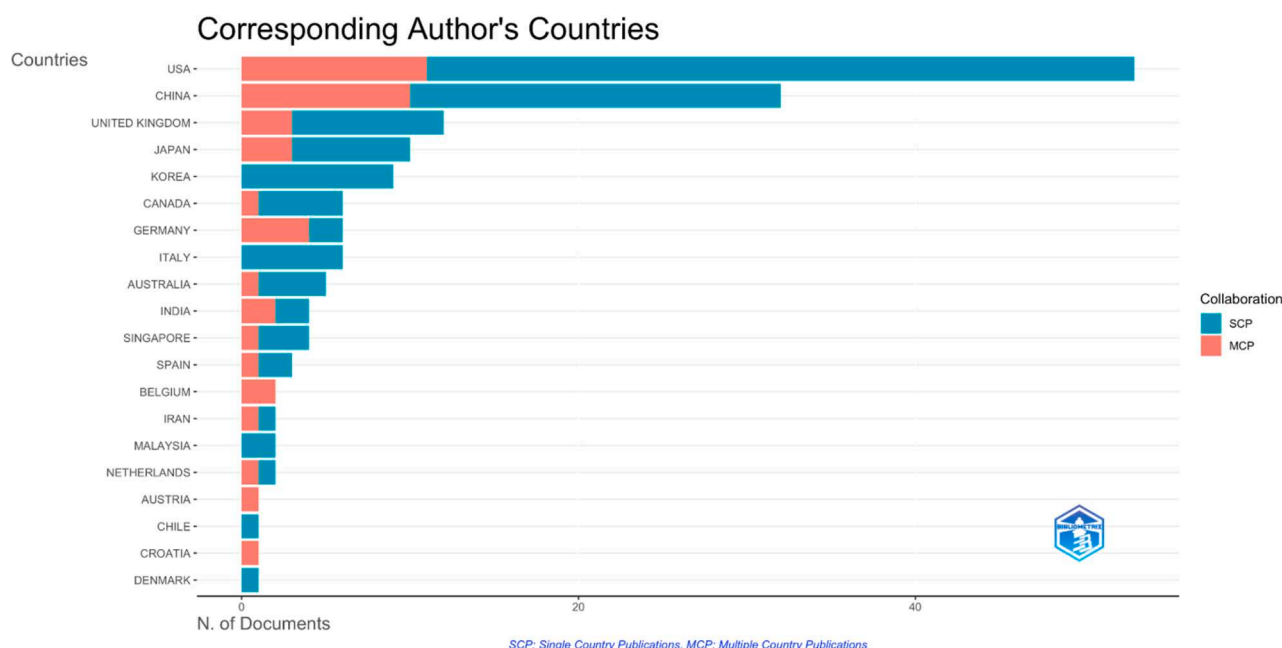


Fig. 3. Analysis of research output.

sclerosis (ALS), as well as aging, indicating a broader, cross-disease perspective supported by regenerative and immune-related concepts. The purple cluster focuses on neural stem cells, and functional outcomes, including cognitive function and depression, as well as preclinical models. Although memory-related and general cognitive terms were dominant, other domains such as executive function, attention, learning ability, cognitive flexibility, behavioral or emotional changes, and motor function were also identified but appeared with lower frequency and weaker co-occurrence strength, resulting in less prominent visualization in the network.

Fig. 7B further demonstrates a temporal shift in research focus, from earlier studies centered on neural stem cells and transplantation toward more recent topics, including extracellular vesicles, neuroinflammation, blood–brain barrier dynamics, and clinical translation. Meanwhile, the density visualization in Fig. 7C highlights Alzheimer's disease, stem cell therapy, and neuroinflammation as the most prominent and intensively studied topics. Overall, these findings indicate a transition from fundamental mechanistic research toward more applied and translational approaches aimed at developing precision regenerative therapies for Alzheimer's disease.

3.7. Strategic diagram

Fig. 8 presents a thematic map based on authors' keywords, organizing research topics along two dimensions: centrality (relevance to the field) and density (level of development). As shown in Fig. 8, the upper-right quadrant represents well-developed and central themes, including AD, neuroinflammation, and stem cells, particularly iPSCs, indicating that these topics form the conceptual foundation and are likely to remain key research priorities. In contrast, the lower-right quadrant includes fundamental but less-developed themes, such as neurodegeneration, regenerative medicine, and apoptosis, which are important to the field but require further theoretical and methodological advancement. The upper-left quadrant highlights highly developed but more specialized or isolated topics, including transplantation, stemness, and mesenchymal stem cells, suggesting strong internal cohesion but weaker connections to broader research areas. Meanwhile, the lower-left quadrant represents emerging or potentially declining themes, such as inflammation and transgenic models, which may reflect either early-stage

development or reduced research attention. Overall, Fig. 9 illustrates a dynamic research landscape that integrates core, well-established themes with specialized domains and emerging concepts in neurodegeneration and stem cell-based therapy. While the thematic map appears to outline predominant research themes, such as neuroinflammation and stem cell-based therapies, specific functional and behavioral domains of Alzheimer's disease are less prominently represented. This reflects the nature of bibliometric mapping, which emphasizes research focus and network connectivity rather than the full multidomain clinical presentation of the disease.

4. Discussion

From 2003 to 2010, Alzheimer's disease research, notably in neuroinflammation and stem cell-based therapeutics, was restricted. Activity increased dramatically after 2016, peaking in 2024 with 29 publications before dropping somewhat in 2025 (Alzheimer's disease facts, 2025). Advances in iPSC and organoid models, as well as the identification of neuroinflammation as a key mechanism, have accelerated efforts (Sun et al., 2025). The peaks in citation impact in 2010 and 2023 indicate that important works continue to receive significant attention despite changes in output.

As measured by publication count, over the last decade, the United States and China have dominated the global research landscape in stem cell-based approaches to AD (Sun et al., 2025). The USA also has the largest network of international collaboration, dollar funding, and multicenter work (Wang et al., 2023). Other nations, such as Germany and India, have produced fewer publications overall, but both feature strong international collaboration. However, research from India is predominantly the product of international cross-border collaborations, with Germany serving as a central hub within the global Alzheimer's disease research network (Liu et al., 2022). In comparison with SCPs, MCPs are more closely associated with increased research productivity and more efficient integration of findings, particularly in international collaboration. The number of publications cited is higher in co-authored studies that address more complex scientific questions, likely because knowledge is shared, leading to a shared methodological foundation (Sun et al., 2025). AD has been moving from descriptive pathology to translational and therapeutic studies (Shi & Yong, 2025). Microglia are

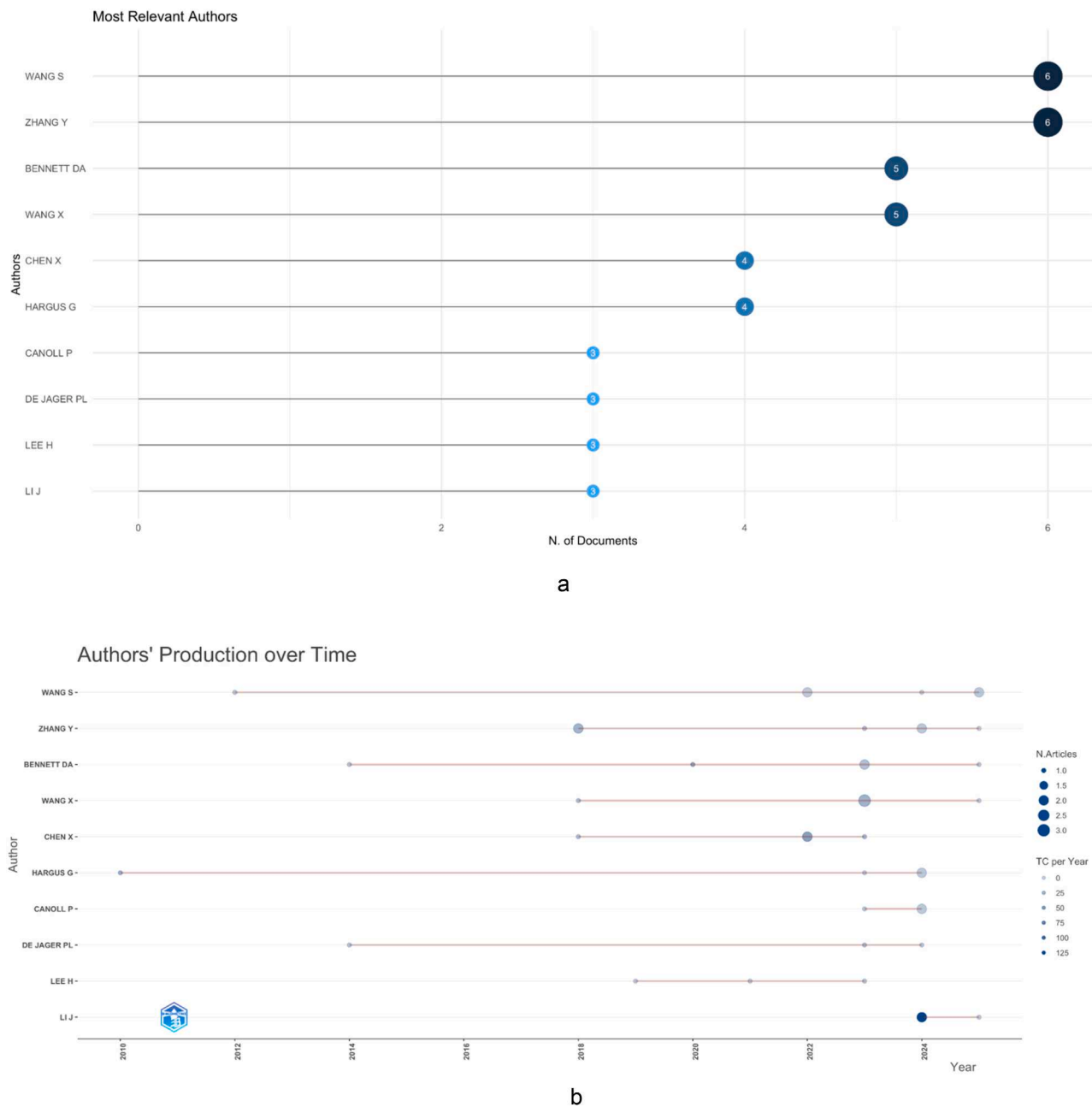


Fig. 4. The most relevant authors.

now considered drivers, rather than responders, of the disease; their activated status by amyloid- β and τ releases cytokines (IL-1, TNF- α), perpetuating the observed neuronal loss (Hansen et al., 2018). The genetic connection, such as TREM2 mutations, also supports the precedence of microglial signaling as a potential therapeutic axis (Wang, 2021).

Meanwhile, regenerative strategies have significantly enhanced the study of AD. Research using mesenchymal and iPSC-derived stem cells has shown promise in improving cognitive function and reducing neuropathological changes in animal models (Ou et al., 2024; Zhang et al., 2020). Exosomes and EVs are also being explored as promising biomarkers and drug carriers across the blood-brain barrier (DeLeo & Ikezu, 2018; Gao et al., 2021). Together, these advances represent a conceptual transformation from classical neurodegeneration, which slows down, towards actual neural restoration.

On a global scale, AD research is accelerating rapidly, with a growing focus on cellular and regenerative sciences. China and the US are the top

two producers and focal points for international cooperation, which has underlain most of the recent dynamics in this area (Wang et al., 2023). Their findings showed that Chinese researchers were up to 11 times more productive. At the same time, US scientists' work was significantly more translational and biomarker-focused, as evidenced by publications in Neurobiology of Aging and Stem Cell Reports. These trends suggest that Alzheimer's research is shifting toward a unified framework in which molecular biology, immune modulation, and regenerative medicine converge to accelerate the translation of experimental results into clinical care.

The transition from descriptive pathology to translational medicine must account for the fact that cultural contexts significantly influence perceptions of AD symptomatology and treatment uptake. Research suggests a fundamental variance in cognitive processing: Westerners tend to be analytical, focusing on objects and logical rules, while East Asians are often holistic, prioritizing contextual information. Consequently, what Western medicine classifies as psychopathology may be

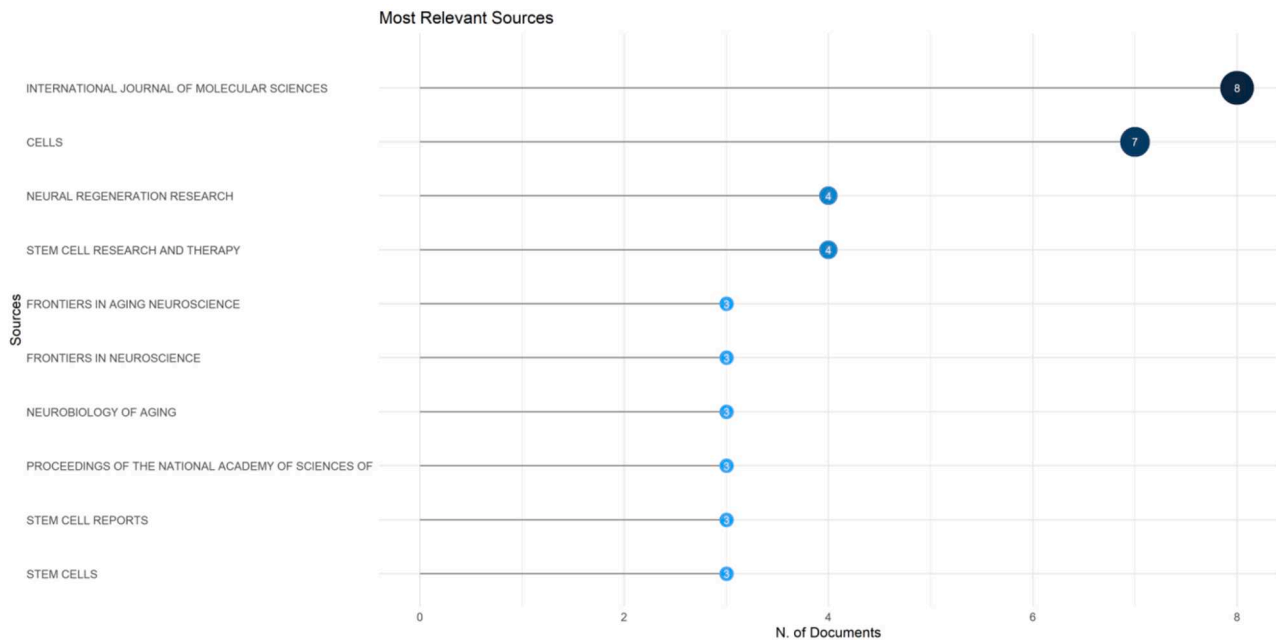


Fig. 5. Based on the number of published.

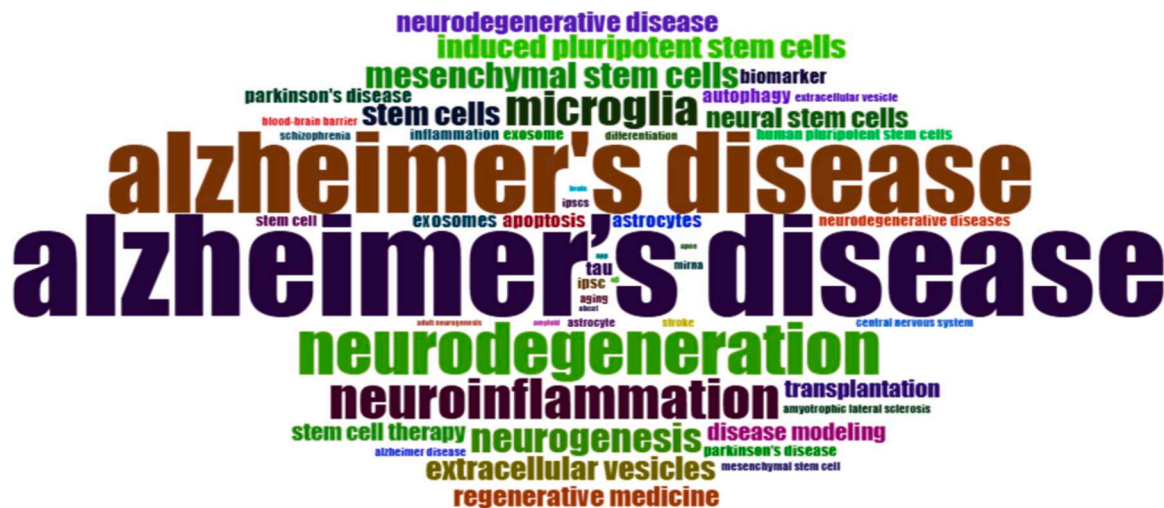


Fig. 6. Bibliomyography analysis.

viewed in some cultures as a natural part of the "normal aging process" or even as a gain in wisdom and authority, as seen in traditional Sub-Saharan societies. These cultural differences clearly affect the reception of pharmaceutical and biological proposals. In cultures where dementia is heavily stigmatized, such as in China, the "cost of failure to seek help" often prevents early intervention. Conversely, countries like Greece prioritize family-centered support in collaboration with health services, which serves as a predictor of quality of life. Even with objective pharmaceutical advances, their uptake may vary significantly, shaping the eventual success of translational strategies across diverse global populations (Feng et al., 2025).

Cluster 1 (in red) highlights the therapeutic potential of mesenchymal stem cell-derived extracellular vesicles (MSC-EVs) as a cell-free strategy for AD. These vesicles are low-immunogenic, with a minimal risk of tumorigenesis, and are capable of crossing the blood-brain barrier (Chen et al., 2021; Yuan & Zhang, 2024). The bioactive molecules, such as proteins, miRNAs, and lipids, carried by MSC-EVs have the potential to downregulate β -amyloid deposition, inhibit the production of

inflammatory cytokines, and modulate microglial activity, which are crucial processes in AD (Kaniowska et al., 2022; Sha et al., 2021). The experimental data indicate that MSC-EVs shield neurons from oxidative and amyloid burdening, along with enhancing cognitive performance as well as regulating inflammation via the Wnt/ β -catenin, I κ B- α , NF- κ Bp65, and NLRP3 inflammasome pathways (Gatti et al., 2020; Sha et al., 2021). However, their clinical translation is hindered by difficulties in EV isolation, dosing, and formulation (Xu et al., 2024). Taken together, MSC-EVs demonstrate excellent safety and multitarget effects, representing a new hope for the treatment of AD.

Cluster 2 (green) underscores the critical role of microglia (CNS immune cells) in neuroinflammation, a process also central to AD, PD, and ALS. In a normal state, microglia play a crucial role in maintaining brain health by clearing cellular debris and supporting neuronal function. However, during chronic activation, they progress to a pro-inflammatory state, characterized by the release of cytokines and ROS, which triggers oxidative stress, neuronal damage, and apoptosis (Wendimu & Hooks, 2022; Yao et al., 2020; Zhao et al., 2026). These

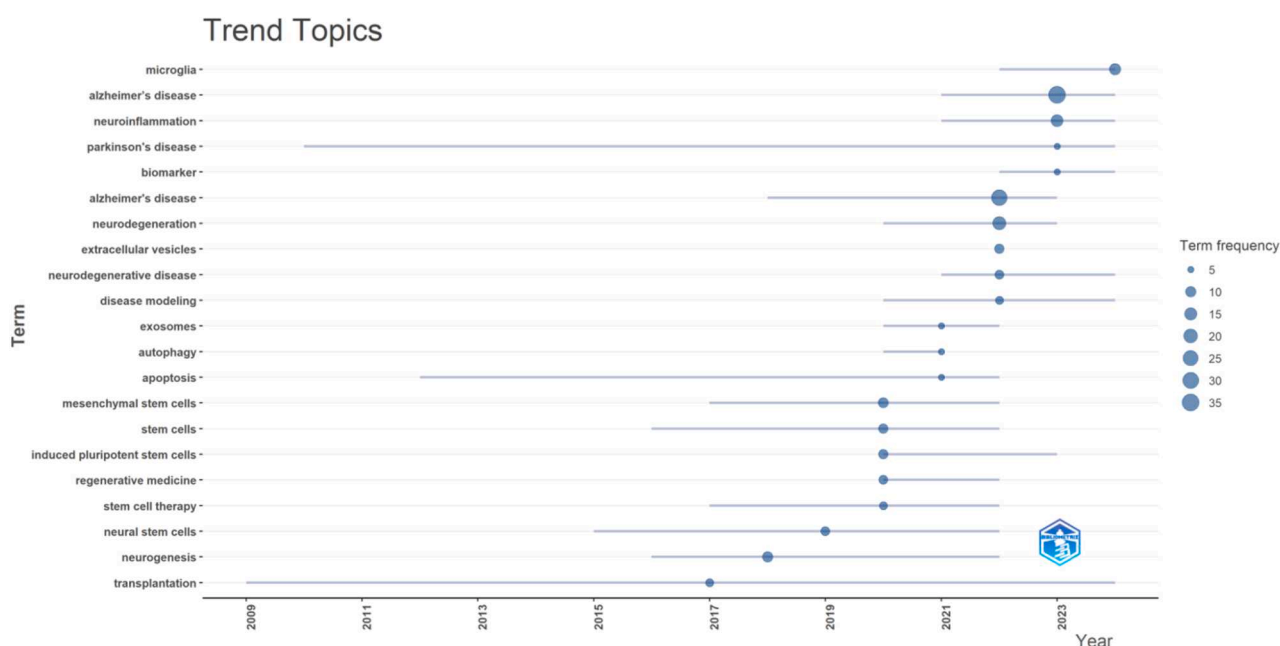


Fig. 7. Trend Topics.

interactions with neurons, astrocytes, the blood–brain barrier, and peripheral immune cells all modulate this inflammatory response (González et al., 2014). Existing therapeutic strategies aim to regulate rather than inhibit microglial activity via pathways such as TREM2 and NF- κ B, or through natural anti-inflammatory compounds such as resveratrol and flavonoids, to manage inflammation without compromising microglial protective functions (Huang et al., 2021).

Cluster 3 (yellow) connects the neurodegeneration supercluster between Parkinson's disease (PD) and amyotrophic lateral sclerosis (ALS) with aging, immune imbalance, and chronic inflammation. The aging process promotes immunosenescence in microglia, which is integrated by T-cells, resulting in persisting activation, oxidative stress, and neuronal loss (Molteni & Rossetti, 2017; Sun et al., 2025). Misfolded proteins, such as α -synuclein in PD and mutant SOD1 in ALS, additionally promote immune activity. Abnormal CD4⁺ T-cell responses drive microglia towards becoming neurotoxic (Gao et al., 2024; Krishnamurthy & Sigurdsson, 2011; Malaspina et al., 2015). Present treatments mainly concentrate on immunomodulation using monoclonal antibodies, and T cell–targeted therapies, as well as regenerative medicine in the form of stem cell therapy, probiotics shifted toward gut–brain balance development, and use of antioxidants such as melatonin to regulate mitochondrial function (Kim et al., 2025; Schwab et al., 2020; Velikic et al., 2024; Verma et al., 2023). In general, this cluster implicates PD and ALS as age-dependent, immune-related neurodegenerative diseases that share inflammatory pathogenesis with AD.

Cluster 4 (purple) NSC transplant has resulted in cognitive and motor improvements in transgenic AD and dementia with Lewy bodies mice, including cases where substantial amyloid reduction does not occur (Blurton-Jones et al., 2009; Zhang et al., 2014). These effects are primarily mediated by enhanced release and activation of brain-derived neurotrophic factor (BDNF), as well as by the MAPK/ERK and PI3K/Akt signaling pathways, which promote neuronal survival and Rapastinel-induced neurogenesis. Therefore, NSC treatment also increases genes associated with neurotransmission, immunity, and mitochondrial metabolism (a more general recovery of the brain Lakatos et al. (2017). Still, local amyloid toxicity can inhibit both NSC migration and differentiation (Li et al., 2018). In addition to memory impairment, AD is characterized by multidomain dysfunction, including deficits in executive function, attention, learning ability, cognitive flexibility, as well as social or emotional behavior and motor function.

These domains are particularly relevant in older populations, where functional decline can substantially impact quality of life. However, such aspects may be underrepresented in bibliometric visualizations, as these analyses prioritize high-frequency keywords and network connectivity rather than the full clinical spectrum of the disease.

The thematic map (Fig. 8), indicating that AD, neuroinflammation, and stem cell research are the front and center subjects of this field, and others, such as regenerative medicine or apoptosis, are still emerging. Specific areas, such as transplantation and stemness, are outside the rim, while other themes, like inflammation and transgenic models, are on the rise. Extracellular vesicle-based therapies also appear promising but require stronger clinical justification (Guo et al., 2020).

The mesenchymal stem cell (MSC) secretome is a potential therapeutic source in AD, owing to its high content of cytokines, growth factors, and extracellular vesicles that promote neuroprotection. Cellular experiments demonstrate that these secretomes stimulate neurogenesis, enhance cognition, and diminish amyloid- β and tau pathology (Reza-Zaldivar et al., 2019), and also subdue microglial and astrocytic activation, thereby promoting the formation of an anti-inflammatory environment conducive to neuronal recovery (Markoutsas et al., 2022). Their mechanism of action involves autophagy-related pathways, such as the PI3K/AKT/mTOR pathway, and could potentially help eliminate toxic proteins (Ebrahim et al., 2024). Noninvasive delivery methods can also yield positive results in animal models, as seen with intranasal cascade delivery (Santamaria et al., 2021).

Nevertheless, inconsistent clinical outcomes highlight the need for standardized manufacturing and robust delivery systems to achieve reliable therapeutic outcomes. Clinical translation remains challenging despite promising data. Prevalent challenges include variable production and purification, which result in differences in composition and breaks in efficacy (Singh et al., 2025); limitations on product scalability; the cost-effectiveness of the processes used; and a lack of well-defined optimal dosing regimens, all of which result in mixed clinical findings (Soares et al., 2022). Second, identifying the active components in secretome mixtures is challenging because these mixtures often contain complex compounds, which can delay regulatory approval. Therefore, although MSC-secretomes have great potential in AD therapy, standardization, a better mechanistic understanding, and scalable manufacturing are urgently needed for their clinical use.

Therapeutic approaches based on the implantation of NSCs and

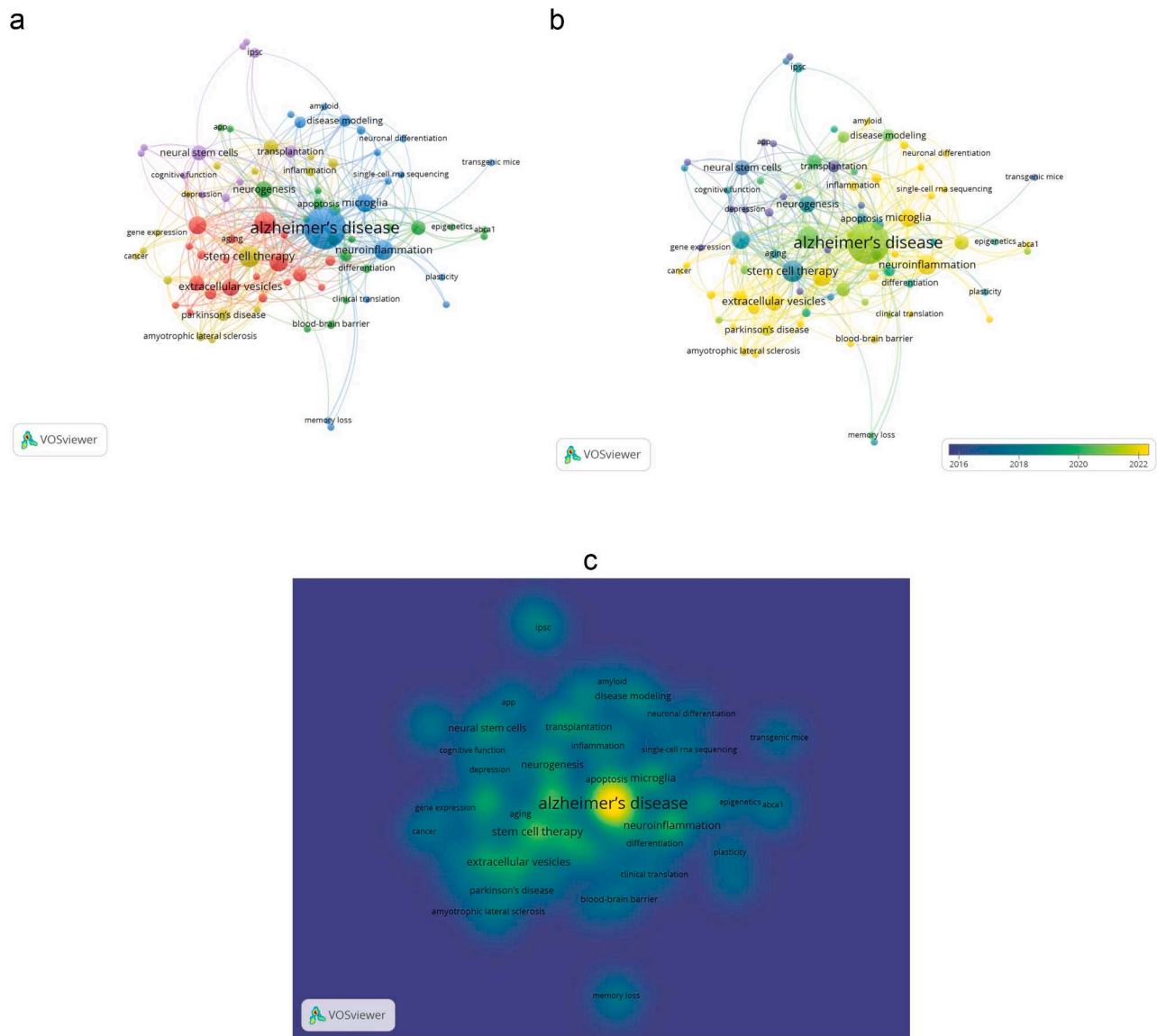


Fig. 8. Co-occurrence network.

iPSCs offer a promising strategy to replace lost neurons and restore neural connections in AD; however, they also face significant translational barriers. Even with improvements in differentiation methods, it remains challenging to control the commitment of pluripotent cells to specific neuronal subsets without inducing unwanted lineages (Chambers et al., 2011). The transplanted cells from both iPSCs and NSCs also carry the risk of being immunogenic and tumorigenic, as they can proliferate in vivo and potentially cause an immune mismatch with the recipients (Dressel, 2011; Gao et al., 2016). Additionally, their therapeutic effect is related to the integration and functional connection of transplanted cells within host neural circuits (Yuan & Zhang, 2024). To ensure autologous application, accurate cell purification, and long-term integration monitoring, these are therefore crucial for clinical safety. Although iPSC technology alleviates some of the ethical concerns associated with embryonic stem cells, it still presents technical challenges in mass production and quality control.

Both cell- and secretome-based therapies encounter ethical, regulatory, and methodological hurdles in their progress towards the clinic. Regulatory agencies have responded slowly, insisting that preclinical validation is rigorous before trial—and that this will be both costly and time-consuming (Trounson et al., 2012). The complex issues of ethical

considerations and concerns associated with cultural and religious contexts also add to the Challenges of Clinical Ethical sensitivity that need to be taken into account during application. Methodologically, the absence of dependable human-like preclinical models and standardized outcome parameters hinders uniform assessment of therapeutic utility (Arber et al., 2017). Secretomes represent a novel avenue in precision medicine, where individual treatments are tailored to an individual's endogenous molecular profile. It is the verbal substance of cell replacement and modulation that contributes to their extraordinary therapeutic effects, but is it secretome-based therapy that attains its benefits via paracrine signaling, immune modulation, and neuroprotection, rendering this type of treatment cell-free (which we know to be less immunogenic than cell-loaded therapies generally) and easier to manufacture/stabilize.

Although cell therapies are regenerative, there are ethical and security concerns regarding their use. Instead, preclinical reports indicate that secretomes can diminish amyloid- β aggregation, tau pathology, and neuroinflammation. However, non-standardized production is considered a significant challenge, primarily because the unstandardized method affects composition and efficacy (Singh et al., 2025). Their clinical application must be confirmed in the same way as drugs,

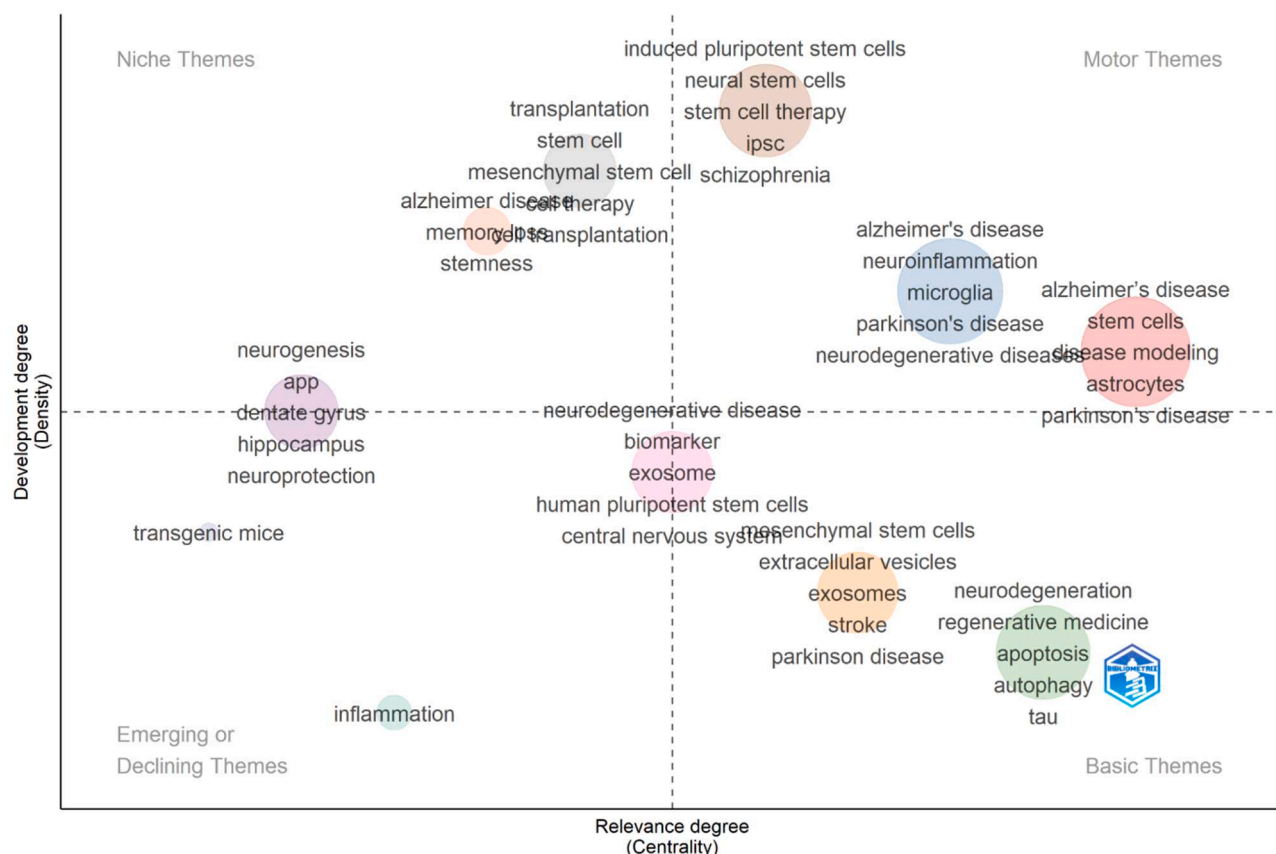


Fig. 9. Thematic map of author keywords.

through rigorous standardization and regulatory approval (Trounson et al., 2012). Advancements in proteomics and bioinformatics are enhancing the characterization of the secretome and the discovery of therapeutic targets (Hsiao et al., 2017).

In summary, secretomes are a safe, scalable, and customizable platform for precision medicine that relies on the combined expertise of multiple disciplines and standardized protocols. Both secretome and stem cell therapies have been applied not only to AD, but also to other neurodegenerative diseases, such as Parkinson's disease and stroke, benefiting the therapy with reduction of neuronal damage or reparation (Kim et al., 2021; Xu et al., 2024). This extended efficacy suggests potential cross-disease significance for AD. Unlike conventional drugs such as cholinesterase inhibitors and memantine, these biological therapies may intervene directly in disease processes and promote the natural regeneration of the disease. Nevertheless, non-drug therapies (cognitive therapy, music therapy, and exercise) are indispensable for cognitive and behavioral functions in AD patients (Liu et al., 2022; Nikkhah Bahrami et al., 2024). When properly combined, conventional therapy and non-pharmacological/alternative/regenerative (NPAR) treatment may provide the most efficacious management for AD. In conclusion, secretomes provide a safe, compatible, and scalable platform for future precision medicine, capable of integrating across multiple disciplines in optimal production systems that can be designed.

The use of a single database (Scopus) may introduce selection bias and limit the inclusion of relevant studies indexed in other databases, such as Web of Science or PubMed. Therefore, the findings of the study should be interpreted with consideration of this limitation. Publication bias and the omission of crucial works, especially from regions or journals underrepresented in Scopus, may have resulted. The depth of themes is also limited by reliance on metadata (titles, abstracts, and keywords), which may neglect more novel or underreported ideas. Additionally, reducing studies to final-stage, English-language journal

publications may introduce bias due to language and selection differences, thereby precluding potentially important non-English or non-journal research. Citation indicators (e.g., total citation and average citation per year) also tend to underestimate the significance of recent work by rewarding the earlier one.

Although VOSviewer and Biblioshiny are successful at visualizing research trends, they do not assess the methodological quality or practical implications of original studies. Thus, these findings must be considered exploratory regarding the therapeutic effects of PGT and should not be interpreted as evidence of its efficacy. Bibliometric analysis can be combined with systematic reviews or meta-analysis to facilitate a comprehensive understanding of clinical outcomes in the future, informed by studies and trials that verify the detected research trends.

5. Conclusion

Recent advances in stem cells and secretome research highlight their growing potential in precision regenerative therapy for AD. The increasing number of publications reflects a global shift toward regenerative and cell-free therapeutic strategies. The secretome derived from MSCs, rich in growth factors, cytokines, and extracellular vesicles, demonstrates significant anti-inflammatory and neuroprotective effects, suggesting a promising alternative to cell-based therapies. Meanwhile, NSCs and iPSCs continue to show potential for neural repair despite ongoing safety, ethical, and technical challenges. The emergence of research clusters focusing on extracellular vesicles, immune modulation, and advanced delivery systems indicates a transition toward more translational and clinically oriented approaches. Future progress will depend on interdisciplinary collaboration, rigorous clinical validation, and the establishment of clear regulatory frameworks to ensure the safe and effective multidomain implementation of personalized therapies for

AD.

Ethics approval and consent to participate

This study used publicly available data; therefore, no additional informed consent or ethics approval is required.

Consent for publication

Not applicable.

Clinical trial number

Not applicable.

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CRedit authorship contribution statement

Tyasno Koeshermanto: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Muhammad Aditya Ferdiansyah:** Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Raden Andi Ario Tedjo:** Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **An'nurihza Zidhan Azhara:** Writing – review & editing, Software, Resources, Project administration, Investigation. **Stefanus Erdana Putra:** Validation, Supervision, Resources, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

This study is based on publicly available bibliographic data from the Scopus database; no primary datasets were generated.

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